

Topic 17: Transition Metals and their Chemistry

Students will be assessed on their ability to:

17.1	know that transition metals are <i>d</i> -block elements that form one or more stable ions with incompletely-filled <i>d</i> -orbitals
17.2	be able to deduce the electronic configurations of atoms and ions of the <i>d</i> -block elements of Period 4 (Sc-Zn) given their atomic number and charge (if any)
17.3	understand why transition metals show variable oxidation number
17.4	know what is meant by the term 'ligand'
17.5	understand that dative (coordinate) covalent bonding is involved in the formation of complex ions
17.6	know that a complex ion is a central metal ion surrounded by ligands
17.7	know that aqueous solutions of transition metal ions are usually coloured
17.8	understand that the colour of aqueous ions, and other complex ions, is a consequence of the splitting of the energy levels of the <i>d</i> -orbitals by ligands
17.9	understand why there is a lack of colour in some aqueous ions and other complex ions
17.10	understand the meaning of the term 'coordination number'
17.11	understand that colour changes in transition metal ions may arise as a result of changes in: - i oxidation number of the ion - ii ligand - iii coordination number of the complex
17.12	understand that H ₂ O, OH ⁻ and NH ₃ act as monodentate ligands
17.13	understand why complexes with six-fold coordination have an octahedral shape, such as those formed by metal ions with H ₂ O, OH ⁻ and NH ₃ as ligands
17.14	know that transition metal ions may form tetrahedral complexes with relatively large ions such as Cl ⁻
17.15	know that square planar complexes are also formed by transition metal ions and that <i>cis</i> -platin is an example of such a complex which is used in cancer treatment where it is supplied as a single isomer and not in a mixture with the <i>trans</i> form
17.16	understand the terms 'bidentate' and 'hexadentate' in relation to ligands, and be able to identify examples such as NH ₂ CH ₂ CH ₂ NH ₂ and EDTA ⁴⁻
17.17	know that haemoglobin is an iron(II) complex containing a polydentate ligand and that ligand exchange occurs when an oxygen molecule bound to haemoglobin is replaced by a carbon monoxide molecule <i>The structure of the haem group will not be assessed.</i>
17.18	know the colours of the oxidation states of vanadium (+5, +4, +3 and +2) in its compounds

17.20	understand, in terms of the relevant E values, that the dichromate(VI) ion, $\text{Cr}_2\text{O}_7^{2-}$ - i can be reduced to Cr^{3+} and Cr^{2+} ions using zinc in acidic conditions - ii can be produced by the oxidation of Cr^{3+} ions using hydrogen peroxide in alkaline conditions (followed by acidification)
17.21	know that the dichromate(VI) ion, $\text{Cr}_2\text{O}_7^{2-}$ can be converted into chromate(VI) ions as a result of the equilibrium $\text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} \rightleftharpoons 2\text{CrO}_4^{2-} + 2\text{H}^+$
17.22	be able to record observations and write suitable equations for the reactions of $\text{Cr}^{3+}(\text{aq})$, $\text{Mn}^{2+}(\text{aq})$, $\text{Fe}^{2+}(\text{aq})$, $\text{Fe}^{3+}(\text{aq})$, $\text{Co}^{2+}(\text{aq})$, $\text{Ni}^{2+}(\text{aq})$, $\text{Cu}^{2+}(\text{aq})$ and $\text{Zn}^{2+}(\text{aq})$ with aqueous sodium hydroxide and aqueous ammonia, including in excess
17.23	be able to write ionic equations to show the meaning of amphoteric behaviour, deprotonation and ligand exchange in the reactions in 17.22
17.24	understand that ligand exchange, and an accompanying colour change, occurs in the formation of: - i $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ from $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ via $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$ - ii $[\text{CuCl}_4]^{2-}$ from $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ - iii $[\text{CoCl}_4]^{2-}$ from $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
17.25	understand, in terms of the positive increase in ΔS_{system} , that the substitution of a monodentate ligand by a bidentate or hexadentate ligand leads to a more stable complex ion
17.26	know that transition metals and their compounds can act as heterogeneous and homogeneous catalysts
17.27	know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction occurs at the surface of the catalyst
17.28	understand, in terms of oxidation number, how V_2O_5 acts as a catalyst in the contact process
17.29	understand how a catalytic converter decreases carbon monoxide and nitrogen monoxide emissions from internal combustion engines by: - i adsorption of CO and NO molecules onto the surface of the catalyst, resulting in the weakening of bonds and chemical reaction - ii desorption of CO_2 and N_2 product molecules from the surface of the catalyst
17.30	know that a homogeneous catalyst is in the same phase as the reactants and appreciate that the catalysed reaction will proceed via an intermediate species
17.31	understand the role of Fe^{2+} ions in catalysing the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$ ions
17.32	know the role of Mn^{2+} ions in autocatalysing the reaction between MnO_4^- and $\text{C}_2\text{O}_4^{2-}$ ions
17.33	CORE PRACTICAL 14 The preparation of a transition metal complex.

	Further suggested practicals: - i prepare the different oxidation states of vanadium (17.18/19) - ii investigate the equilibrium reaction in 17.21 - iii carry out the reactions of transition metal ions with sodium hydroxide and ammonia solutions (17.22) - iv carry out the ligand exchange reactions in 17.24 - v investigate the kinetics of the reaction between MnO_4^- and $\text{C}_2\text{O}_4^{2-}$ in 17.32
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